One Rule Instead of Five: A Binomial Mnemonic for Teaching Moment Transformations

Abstract

Converting between raw and central moments is taught repeatedly across the statistics curriculum—from higher-secondary syllabi through graduate distribution theory—and is, at every level, taught in much the same unrewarding way: as four or five separate, order-specific algebraic formulas to be memorized or looked up, with no visible structure connecting them. This paper describes a teaching approach that replaces the whole collection with a single rule. The underlying identity—that an origin shift from a to k acts on moments as a binomial expansion in $b = a - k$—is classical and appears in the advanced literature (Kendall and Stuart 1977), but it is rarely taught as the organizing principle it is. We present it as a symbolic mnemonic: write $(a + b)^r$, expand it with Pascal's triangle, and read a^j as "the j-th moment about a." One device then covers raw-to-raw, raw-to-central, and central-to-raw conversion at any order, for sample data and for named distributions alike. Because the rule requires only the binomial theorem and linearity of expectation, it is usable from secondary level upward without modification. We report classroom experience with higher-secondary students (classes 11–12), who applied it successfully despite having no calculus-based alternative available—the most demanding audience for the material rather than the narrowest. An accompanying R package, `convmoment`, implements the rule directly and lets students check derivations against software rather than against an answer key.

Keywords: statistics education, moment transformation, central moments, mnemonics, R, mathematical statistics

1 Introduction

1.1 Why Moment Transformations Are Harder to Teach Than They Should Be

Moments are introduced early in mathematical statistics. Raw moments about an arbitrary origin a,

$$\mu_r'(a) = \mathbb{E}[(X - a)^r],$$

and central moments about the mean μ,

$$\mu_r = \mathbb{E}[(X - \mu)^r],$$

carry the structural information behind location, scale, skewness, and kurtosis, and moving between them is a routine operation. The topic recurs across the curriculum at very different levels of sophistication. It appears in secondary and higher-secondary syllabi wherever moments, skewness, and kurtosis are taught descriptively; it reappears in undergraduate distribution theory in expectation form; and it is used without comment in graduate work whenever variance is obtained from $\mathbb{E}[X^2]$, whenever skewness or kurtosis is formed, or whenever moment-based estimators are derived.

That recurrence matters for what follows. The conversion is taught more than once, to audiences with very different mathematical equipment, and—as we argue below—it is taught badly in much the same way each time.

The topic is elementary, but in the author's experience it teaches poorly. The standard presentation gives a list:

$$\mu_2 = \mu_2'(a) - [\mu_1'(a)]^2$$

$$\mu_3 = \mu_3'(a) - 3\mu_2'(a)\mu_1'(a) + 2\,[\mu_1'(a)]^3$$

$$\mu_4 = \mu_4'(a) - 4\mu_3'(a)\mu_1'(a) + 6\mu_2'(a)\,[\mu_1'(a)]^2 - 3\,[\mu_1'(a)]^4$$

$$\begin{aligned}\mu_5 = \mu_5'(a) - 5\mu_4'(a)\mu_1'(a) + 10\mu_3'(a)\,[\mu_1'(a)]^2 \\ - 10\mu_2'(a)\,[\mu_1'(a)]^3 + 4\,[\mu_1'(a)]^5 .\end{aligned}$$

Nothing in this display tells a student where the coefficients $3, 2$ or $4, 6, 3$ or $5, 10, 10, 4$ come from, why the signs alternate, or how to write down the next line. The formulas are typically presented in this form in widely used texts (Cramér 1946; Mood et al. 1974; Johnson et al. 1994). A student who forgets one has no route back to it except a table. A student who needs μ_6, or who needs to shift to an origin that is not the mean, is simply stuck.

This is a familiar failure mode: a set of results that are individually correct, collectively unmemorable, and presented without the structure that generates them. The predictable consequences are that students treat moment conversion as lookup rather than derivation, that algebra errors in the higher-order expansions go undetected, and that the connection between "shifting the origin" and "re-expanding a polynomial" is never made explicit.

1.2 What This Paper Does—and Does Not—Claim

We describe a way of teaching this material in which the five formulas above are replaced by one rule that generates all of them, at any order, for any pair of origins.

We want to be clear at the outset about the status of the underlying mathematics. **The identity is classical.** That a change of origin acts on moments through a binomial expansion follows immediately from the binomial theorem, and the relation appears in the standard advanced reference literature (Kendall and Stuart 1977) and is implicit in any treatment that derives the conversion formulas at all. It is also what general-purpose moment software computes internally. We claim no new theorem, and the paper should not be read as offering one.

The contribution is pedagogical, and consists of three parts:

1. **A mnemonic that makes the rule usable at the board.** Recasting the identity as the symbolic expansion $(a+b)^r$ with the substitution $a^j \mapsto \mu'_j(a)$ turns a formula students must recall into a procedure they can execute with Pascal's triangle. This framing, rather than the identity itself, is what makes the material teachable.

2. **A single organizing parameter.** Introducing the origin shift $b = a - k$ collapses what textbooks present as three distinct topics—raw-to-raw, raw-to-central, and central-to-raw conversion—into one operation with three choices of b. Students learn one thing instead of three.

3. **Software that mirrors the derivation.** The R package `convmoment` implements the rule in the same form in which it is taught, so students and educators can verify their own hand derivations at any order rather than checking against printed answers.

Section 2 sets out the standard treatment and diagnoses precisely where it loses students. Section 3 develops the single rule and the mnemonic. Section 4 reports classroom experience. Section 5 shows the same device applied to named distributions, where it replaces repeated differentiation of moment generating functions. Section 6 describes the R package and how it is used in assignments. Section 7 discusses what the approach does not fix, including a numerical caution that is itself worth teaching.

1.3 Why the Approach Is Level-Independent

The proposal is deliberately not tied to a particular stage of the curriculum, and it is worth saying why, since the argument is structural rather than merely optimistic.

The rule has exactly two prerequisites: the binomial expansion of $(a + b)^r$, and the fact that expectation—or, at a more elementary level, averaging—is linear. Both are available well before moments are introduced. The binomial theorem is standard secondary-school algebra, and Pascal's triangle is typically the form in which students first meet it. Nothing in the rule requires calculus, generating functions, or measure-theoretic expectation. The ceiling is equally open: the same expansion is what one wants at graduate level, where it composes with cumulant machinery and extends to the multivariate case.

This gives the approach an unusual property for a teaching device. It is not a simplification that must be unlearned later, and it is not an advanced technique retrofitted downward. It is the same rule throughout, and a student who meets it at secondary level meets the object they will keep using. What changes with level is only the surrounding motivation and how far the rule is pushed—not the rule.

The classroom experience we report in Section 4 comes from the lower end of that range, with students at higher-secondary level. We regard that as the more demanding test rather than a narrower one: it is the audience with the least mathematical equipment, and an approach that succeeds there has cleared the harder bar. We do not, however, present it as evidence about undergraduate or graduate instruction, where the argument above is structural and untested.

1.4 Where the Skill Is Needed Later

It is worth telling students that this is not a self-contained exercise, though the motivation an instructor reaches for will depend on the audience.

For students meeting moments descriptively, the persuasive case is usually internal to the syllabus: skewness and kurtosis are defined through central moments, computation is almost always easier about a convenient origin than about the mean, and the origin shift is what connects the two. In our own teaching this—rather than any downstream application—is what makes the rule feel worth having.

For more advanced audiences, origin shifts on moments recur throughout applied statistics: in Edgeworth and Gram–Charlier expansions, where higher-order central moments must be converted to standardized moments or cumulants (Hall 1992; Blinnikov and Moessner 1998); in the generalized method of moments, where sample moments computed about convenient origins are matched to population moments (Hansen 1982); and in image analysis, where moment invariants are obtained by shifting the spatial origin to the centroid (Hu 1962; Mukundan and Ramakrishnan 1998). The last of these makes an especially direct point: translation invariance *is* an origin shift.

2 The Standard Treatment and Where It Fails

2.1 The Derivation as Usually Presented

Consider transforming raw moments from an initial origin a to a target origin k. The textbook route writes $(x_i - k)$ as $[(x_i - a) + (a - k)]$, expands for the specific order required, and distributes the summation.

2.1.1 A Worked Example

Suppose the first three moments of a dataset about origin $a = 2$ are

$$\mu_1'(2) = -1, \quad \mu_2'(2) = 7, \quad \text{and} \quad \mu_3'(2) = 39,$$

and the corresponding moments about $k = 5$ are wanted.

2.1.1.1 First moment ($r = 1$)

$$\begin{aligned}\mu_1'(5) &= \frac{1}{n}\sum_{i=1}^{n}(x_i - 5) = \frac{1}{n}\sum_{i=1}^{n}[(x_i - 2) - 3] \\ &= \frac{1}{n}\sum_{i=1}^{n}(x_i - 2) - 3 = \mu_1'(2) - 3 = -1 - 3 = -4.\end{aligned}$$

2.1.1.2 Second moment ($r = 2$)

$$\begin{aligned}\mu_2'(5) &= \frac{1}{n}\sum_{i=1}^{n}(x_i - 5)^2 = \frac{1}{n}\sum_{i=1}^{n}[(x_i - 2) - 3]^2 \\ &= \frac{1}{n}\sum_{i=1}^{n}\Big[(x_i - 2)^2 - 6(x_i - 2) + 9\Big] \\ &= \mu_2'(2) - 6\mu_1'(2) + 9 = 7 - 6(-1) + 9 = 22.\end{aligned}$$

2.1.1.3 Third moment ($r = 3$)

$$\begin{aligned}\mu_3'(5) &= \frac{1}{n}\sum_{i=1}^{n}(x_i - 5)^3 = \frac{1}{n}\sum_{i=1}^{n}[(x_i - 2) - 3]^3 \\ &= \frac{1}{n}\sum_{i=1}^{n}\Big[(x_i - 2)^3 - 9(x_i - 2)^2 + 27(x_i - 2) - 27\Big] \\ &= \mu_3'(2) - 9\mu_2'(2) + 27\mu_1'(2) - 27 \\ &= 39 - 9(7) + 27(-1) - 27 = -78.\end{aligned}$$

2.2 Diagnosing the Difficulty

Worked at the board, this is unobjectionable. The difficulty is what students take away from it.

First, **the structure is present but invisible**. The coefficients $1, -6, 9$ and $1, -9, 27, -27$ are binomial coefficients times powers of the shift, but because the shift is buried inside the summation and the expansion is done afresh each time, students see three unrelated computations rather than one pattern instantiated three times. In our experience, asking a class what $r = 4$ would look like after this presentation produces hesitation, not extrapolation.

Second, **the cost grows with order while the insight does not**. Deriving the case $r \geq 4$ by hand means expanding a higher-degree polynomial, and the probability of an algebra slip rises accordingly. Students spend their effort on expansion mechanics, which is not what the topic is meant to teach.

Third, **the special cases look unrelated**. Conversion to central moments is presented as its own topic with its own formulas (the display in Section 1), rather than as the same operation with the origin chosen to be $\bar{x}$. A student who has learned both has learned two things and knows of no reason they should be connected.

The approach in the next section addresses all three: it exposes the pattern, makes the cost of higher orders negligible, and makes the special cases visibly special cases.

3 One Rule for Every Conversion

3.1 The Identity

Let a be the initial origin and k the target origin, and define the **origin shift**

$$b = a - k.$$

For any random variable X,

$$X - k = (X - a) + (a - k) = (X - a) + b,$$

so raising to the power $r \in \mathbb{N}$ and applying the binomial theorem gives

$$(X-k)^r = \sum_{j=0}^{r} \binom{r}{j} (X-a)^j b^{r-j},$$

and taking expectations term by term,

$$\mathbb{E}[(X-k)^r] = \sum_{j=0}^{r} \binom{r}{j} \mathbb{E}[(X-a)^j] b^{r-j}.$$

With $\mathbb{E}[(X-a)^j] = \mu'_j(a)$ and the convention $\mu'_0(a) \equiv 1$:

The origin-shift rule.

For any initial origin a, target origin k, and origin shift $b = a - k$,

$$\mu'_r(k) = \sum_{j=0}^{r} \binom{r}{j} \mu'_j(a)\, b^{r-j}.$$

As noted in Section 1, this is a classical result rather than a new one; see Kendall and Stuart (1977) for the standard reference treatment. The derivation above is three lines and uses only the binomial theorem and linearity of expectation, both of which students already have. That it is short is the point: it can be re-derived at the board in the time it takes to look up a table, which is what makes it safe for students to stop memorizing the special cases.

3.2 The Mnemonic

The rule as displayed is still a summation formula, and summation formulas are not what students reach for under exam conditions. The device we actually teach is the symbolic substitution

$$\mu'_r(k) \equiv (a+b)^r \quad \text{with } a^j \mapsto \mu'_j(a) \text{ and } a^0 \equiv 1.$$

That is: *write down the binomial expansion of $(a+b)^r$ exactly as in secondary-school algebra, then read every power a^j as "the j-th moment about a."* The coefficients come from Pascal's triangle, which students already know and can regenerate if they forget. Or, they can also use $\binom{r}{i}$, where r is the order of the moment, and i represents ith term of binomial expansion.

Thus, using the expression $\binom{r}{i}$, the coefficients, for example, second-order binomial expansion, the coefficeints are $\binom{2}{2}$, $\binom{2}{1}$, $\binom{2}{0}$, which are calculated as 1, 2, and 0, respectively. Higher-order coefficients can be found similarly.

Table 1 shows the result for orders 1 through 5.

Moment Order (r)	Symbolic Binomial Form	Explicit Moment Formula ($\mu'_r(k)$)
1	$(a+b)^1$	$\mu'_1(a)+b$
2	$(a+b)^2$	$\mu'_2(a)+2\mu'_1(a)b+b^2$
3	$(a+b)^3$	$\mu'_3(a)+3\mu'_2(a)b+3\mu'_1(a)b^2+b^3$
4	$(a+b)^4$	$\mu'_4(a)+4\mu'_3(a)b+6\mu'_2(a)b^2+4\mu'_1(a)b^3+b^4$
5	$(a+b)^5$	$\mu'_5(a)+5\mu'_4(a)b+10\mu'_3(a)b^2+10\mu'_2(a)b^3+5\mu'_1(a)b^4+b^5$

Table 1: The symbolic mnemonic for moment orders 1–5. Students write the middle column from Pascal's triangle and translate it into the right column by reading a^j as $\mu'_j(a)$.

The coefficients are read directly from Pascal's triangle:

$$
\begin{array}{c}
1 \\
1 \quad 1 \\
1 \quad 2 \quad 1 \\
1 \quad 3 \quad 3 \quad 1 \\
1 \quad 4 \quad 6 \quad 4 \quad 1 \\
1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1
\end{array}
$$

A point worth making explicitly in class: the alternating signs that make the textbook central-moment formulas look arbitrary are not part of the rule at all. They appear because the raw-to-central case has $b = -\mu_1'(a)$, so odd powers of b carry a minus sign. Once students see that, the sign pattern stops being something to remember.

3.3 Three Conversions, One Operation

The single parameter $b = a - k$ covers every case students meet:

1. **Raw-to-raw.** $b = a - k$ for arbitrary origins a and k.
2. **Raw-to-central.** The target origin is the mean, $k = \bar{x} = \mu_1'(a) + a$, so $b = a - (\mu_1'(a) + a) = -\mu_1'(a)$. When the initial origin is zero this is just $b = -\bar{x}$.
3. **Central-to-raw.** The initial origin is the mean, $a = \bar{x}$, and the target is $k = 0$, so $b = \bar{x}$.

Presenting these as a single slide, immediately after the mnemonic, is where the approach earns its keep: three chapters' worth of formulas reduce to three choices of one parameter.

3.4 The Example Redone

Returning to $a = 2$, $k = 5$, with $\mu_1'(2) = -1$, $\mu_2'(2) = 7$, $\mu_3'(2) = 39$, we have $b = 2-5 = -3$ and using the corresponding binomial forms,

1. $\mu_1'(5) = \mu_1'(2) + b = -1 + (-3) = -4$.
2. $\mu_2'(5) = \mu_2'(2) + 2\mu_1'(2)b + b^2 = 7 + 2(-1)(-3) + (-3)^2 = 7 + 6 + 9 = 22$.
3. $\mu_3'(5) = \mu_3'(2) + 3\mu_2'(2)b + 3\mu_1'(2)b^2 + b^3 = 39 + 3(7)(-3) + 3(-1)(-3)^2 + (-3)^3 = 39 - 63 - 27 - 27 = -78$.

The answers match Section 2, as they must, but no polynomial was expanded and no summation was distributed. Extending to $r = 6$ requires reading one more row of Pascal's triangle or one can simply use the expression $\binom{r}{i}$, as shown before.

3.5 The Matrix Form

For students who have had linear algebra, the rule has a compact matrix statement that is worth showing as a coda. Writing $\boldsymbol{\mu}'(a) = [1, \mu_1'(a), \mu_2'(a), \ldots, \mu_K'(a)]^T$ for the vector of moments up to order K,

$$\boldsymbol{\mu}'(k) = \mathbf{C}_b\, \boldsymbol{\mu}'(a), \qquad [\mathbf{C}_b]_{r,j} = \begin{cases} \dbinom{r}{j} b^{r-j} & \text{for } 0 \le j \le r \le K, \\ 0 & \text{otherwise,} \end{cases}$$

so $\mathbf{C}_b$ is lower triangular of dimension $(K+1) \times (K+1)$. Two facts follow immediately and make good exercises: $\mathbf{C}_b^{-1} = \mathbf{C}_{-b}$, so inverting a conversion means negating the shift; and $\mathbf{C}_{b_1}\mathbf{C}_{b_2} = \mathbf{C}_{b_1+b_2}$, so successive origin shifts compose additively. Neither is obvious from the textbook formula lists.

This subsection is intended for audiences that have had linear algebra and is not part of the minimum presentation; it is omitted entirely in the secondary-level teaching described in Section 4. It is included here because it shows what the rule becomes at a higher level—the same content, restated—which is the level-independence claim of Section 1.3 made concrete.

4 Classroom Experience

4.1 Setting

The approach has been used by the author in higher-secondary teaching (classes 11–12) at a residential secondary institution in Bangladesh,[1] within the first Statistics paper, in the chapter on moments, skewness, and kurtosis. Each cohort was approximately 30 students, and the approach has been used with successive cohorts over six years.

The prerequisites the rule needs were already in place. Students had met the binomial theorem and Pascal's triangle in algebra, and were familiar with expectation and variance. They had not previously been shown the origin-shift rule; the conventional treatment in the prescribed textbook is the formula list reproduced in Section 1.

This is the least-equipped audience the material is normally taught to, which is why we regard it as an informative test rather than a limited one (Section 1.3). In particular, these students had no calculus-based route to the same results: the generating-function derivations of Section 5 were unavailable to them, so the origin-shift rule was not a shortcut past a method they already had—it was the only method they had.

The material was delivered in a single class session, structured as in Sections 2 and 3: the conventional derivation first, then why the formula list is hard to retain, then the rule and the mnemonic, then the three cases of b. The slides used are described in Section 11; the instruction they give students, stated on the slide itself, is simply *do not memorize.*

[1]Institution named in the non-anonymized version.

4.2 What Students Did

At first they were shown the traditional textbook approach to change origin of raw moments. Then they were told that changing to central moments from raw moments required a different set of formulae. Then the new method was introduced. But before that, the students were first reintroduced to the binomial expansion and how to obtain its coefficients with the help of Pascal's triangle and the expression $\binom{r}{i}$. They were then asked to associate the rth moment with the rth binomial expansion.

The author's observation was that students applied the transformation without substantial difficulty. Becoming familiar with the notation took a short period—typically no more than five minutes—after which they proceeded with ease, converting the first four moments successfully.

We are careful about what this does and does not show. Students applied the rule correctly at the orders covered in the session; we did not set a task at an order withheld from instruction, which would have been the sharper test of whether the mnemonic transfers rather than merely supports a worked example. The observation is therefore that the rule was learnable and usable in one session by students with only algebraic prerequisites—not that it was demonstrably generative beyond the orders taught. Section 7 returns to this.

We state plainly that this is informal classroom experience, not a study. There was no comparison group, no randomization, no validated instrument, and the observations are the instructor's own and were not recorded for research purposes at the time. We report the experience because it was consistent and because the mechanism is plausible on its face: the mnemonic replaces recall of several unrelated formulas with execution of one procedure students already possess. We would not want it read as an effect size, and Section 7 says what a proper evaluation would require.

4.3 What Gave Students Trouble

One difficulty was consistent enough to be worth reporting. Students stumbled when moving from raw moments to central moments, and the sticking point was not the expansion but the origin: they were unsure where the new origin was supposed to come from. The mean is not given separately in this setting, and students did not initially see that it is already available from the moments they hold, as $\bar{x} = \mu_1'(a) + a$. Once it was shown that the target origin can be derived from the raw moments themselves—the raw-to-central case of Section 3.3—the difficulty resolved.

This is worth flagging for anyone adopting the approach, because it is a predictable consequence of how the rule is presented. In the raw-to-raw case b is a constant the student chooses; in the raw-to-central case $b = -\mu_1'(a)$ depends on the data. Students who have met only the first case reasonably expect the second to supply an origin from outside. Making that dependence explicit when the three cases are introduced, rather than leaving it implicit in the algebra, appears to prevent the confusion rather than merely fix it afterwards.

5 The Same Rule for Named Distributions

A useful feature of the rule for teaching is that it is not a data-handling trick: it applies unchanged to theoretical distributions, where it replaces repeated differentiation of moment generating functions. Showing this immediately after the sample-data case reinforces that moments are properties of distributions, not of datasets.

5.1 The Binomial Distribution

Let $X \sim \text{Bin}(m, p)$; we write the number of trials as m rather than n to keep it distinct from sample size. The moment generating function is $M_X(t) = (1 - p + pe^t)^m$, and differentiating it at $t = 0$ gives the raw moments in Table 2. The central moments in the same table are the target, and all four are derived below. To keep the raw moments compact we write the falling factorial

$$m^{(j)} = m(m-1)(m-2)\cdots(m-j+1),$$

so that $m^{(1)} = m$, $m^{(2)} = m(m-1)$, and so on.

Order (r)	Raw moment $\mu_r'(0)$	Central moment μ_r
1	$m^{(1)}p$	0
2	$m^{(1)}p + m^{(2)}p^2$	$mp(1-p)$
3	$m^{(1)}p + 3m^{(2)}p^2 + m^{(3)}p^3$	$mp(1-p)(1-2p)$
4	$m^{(1)}p + 7m^{(2)}p^2 + 6m^{(3)}p^3 + m^{(4)}p^4$	$mp(1-p)\left[1 + 3(m-2)p(1-p)\right]$

Table 2: Raw and central moments of the Binomial distribution $\text{Bin}(m, p)$ through order four. The raw moments follow from the moment generating function, written here in falling-factorial form for compactness ($m^{(j)}$ defined above); the central moments are derived from them in Section 5.1.2 using the origin-shift rule alone.

Expanding the order-two and order-three entries of the raw column gives the forms used explicitly in the derivations below:

$$\mu_2'(0) = mp(1-p) + m^2p^2, \qquad \mu_3'(0) = mp - 3mp^2 + 3m^2p^2 + 2mp^3 - 3m^2p^3 + m^3p^3.$$

Setting the raw moments out in advance, rather than producing them as they are needed, is deliberate: it makes visible that the origin shift consumes exactly this table and nothing else. Two features are worth pointing out to students before any algebra begins. The central moments are dramatically simpler than the raw ones—compare $mp(1-p)(1-2p)$ with the six-term expansion of $\mu_3'(0)$—which is the practical reason the conversion is worth having. And the fully expanded raw moments grow unmanageable quickly: at order four the expansion runs to ten terms, which is why the falling-factorial form is used in the table. The origin-shift rule is indifferent to this; it consumes whichever form is supplied.

5.1.1 The MGF Route to Central Moments

To obtain central moments directly from the generating function, one differentiates $M_{X-\mu}(t) = e^{-mpt}(1-p+pe^t)^m$. For μ_2 this is manageable. For μ_3 it requires the third derivative of a product of two functions, each of which is itself a composition—three applications of the product rule with the chain rule inside each, evaluated at $t = 0$. The algebra is several lines long and is, in our experience, where students abandon the derivation.

5.1.2 The Origin-Shift Route

The rule needs only the raw moments already tabulated and the shift. Here $a = 0$, $k = \mu = mp$, so $b = -mp$.

1. **First central moment.** $\mu_1 = \mu_1'(0) + b = mp - mp = 0$.

2. **Second central moment (the variance).**

$$\begin{aligned}\mu_2 &= \mu_2'(0) + 2\mu_1'(0)b + b^2 \\ &= \left[mp(1-p) + m^2p^2\right] + 2(mp)(-mp) + (-mp)^2 \\ &= mp - mp^2 + m^2p^2 - 2m^2p^2 + m^2p^2 = mp(1-p).\end{aligned}$$

3. **Third central moment.** Taking $\mu_3'(0)$ from Table 2,

$$\begin{aligned}
\mu_3 &= \mu_3'(0) + 3\mu_2'(0)b + 3\mu_1'(0)b^2 + b^3 \\
&= \left(mp - 3mp^2 + 3m^2p^2 + 2mp^3 - 3m^2p^3 + m^3p^3\right) \\
&\quad + 3\left(mp - mp^2 + m^2p^2\right)(-mp) + 3(mp)(-mp)^2 + (-mp)^3 \\
&= mp(1-p)(1-2p).
\end{aligned}$$

4. **Fourth central moment.** With $b = -mp$ the rule gives

$$\begin{aligned}
\mu_4 &= \mu_4'(0) + 4\mu_3'(0)b + 6\mu_2'(0)b^2 + 4\mu_1'(0)b^3 + b^4 \\
&= \mu_4'(0) - 4mp\,\mu_3'(0) + 6m^2p^2\,\mu_2'(0) - 4m^3p^3\,\mu_1'(0) + m^4p^4 \\
&= mp(1-p)\left[1 + 3(m-2)p(1-p)\right],
\end{aligned}$$

where the last step is routine collection of terms. Writing down the substitution is immediate; only the simplification is laborious, and that labour belongs to the distribution rather than to the method.

Each central moment of order r draws on the tabulated raw moments of orders 1 through r, and on nothing else—the same triangular dependence recorded by the lower-triangular matrix $\mathbf{C}_b$ of Section 3.5. Every entry of the central column of Table 2 has now been recovered from the raw column, using no property of the Binomial distribution beyond the raw moments themselves.

5.2 The Normal Distribution

Table 3 gives the raw and central moments of $X \sim \mathcal{N}(\mu, \sigma^2)$ through order four.

5.2.1 Raw to Central ($b = 0 - \mu = -\mu$)

1. $\mu_1 = \mu_1' + b = \mu - \mu = 0.$

Order (r)	Raw moment $\mu_r'(0)$	Central moment μ_r
1	μ	0
2	$\mu^2 + \sigma^2$	σ^2
3	$\mu^3 + 3\mu\sigma^2$	0
4	$\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$	$3\sigma^4$

Table 3: Raw and central moments of the Normal distribution through order four.

2. $\mu_2 = \mu_2' + 2\mu_1' b + b^2 = (\mu^2 + \sigma^2) + 2\mu(-\mu) + (-\mu)^2 = \sigma^2$.

3. Third:

$$\begin{aligned}\mu_3 &= \mu_3' + 3\mu_2' b + 3\mu_1' b^2 + b^3 \\ &= (\mu^3 + 3\mu\sigma^2) + 3(\mu^2 + \sigma^2)(-\mu) + 3\mu(-\mu)^2 + (-\mu)^3 \\ &= \mu^3 + 3\mu\sigma^2 - 3\mu^3 - 3\mu\sigma^2 + 3\mu^3 - \mu^3 = 0.\end{aligned}$$

4. Fourth:

$$\begin{aligned}\mu_4 &= \mu_4' + 4\mu_3' b + 6\mu_2' b^2 + 4\mu_1' b^3 + b^4 \\ &= (\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4) + 4(\mu^3 + 3\mu\sigma^2)(-\mu) + 6(\mu^2 + \sigma^2)(-\mu)^2 + 4\mu(-\mu)^3 + (-\mu)^4 \\ &= (\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4) - (4\mu^4 + 12\mu^2\sigma^2) + (6\mu^4 + 6\mu^2\sigma^2) - 4\mu^4 + \mu^4 \\ &= 3\sigma^4.\end{aligned}$$

5.2.2 Central to Raw ($b = \mu - 0 = \mu$)

Running the same rule in the other direction, with $a = \mu$, $k = 0$, $b = \mu$:

$$\mu_1'(0) = \mu_1 + b = 0 + \mu = \mu,$$

$$\mu_2'(0) = \mu_2 + 2\mu_1 b + b^2 = \sigma^2 + 0 + \mu^2 = \mu^2 + \sigma^2,$$

$$\mu_3'(0) = \mu_3 + 3\mu_2 b + 3\mu_1 b^2 + b^3 = 0 + 3\sigma^2\mu + 0 + \mu^3 = \mu^3 + 3\mu\sigma^2,$$

$$\mu_4'(0) = \mu_4 + 4\mu_3 b + 6\mu_2 b^2 + 4\mu_1 b^3 + b^4 = 3\sigma^4 + 0 + 6\sigma^2\mu^2 + 0 + \mu^4 = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4.$$

That the same device runs in both directions, with only the sign of b changed, is worth pausing on in class—it is the $\mathbf{C}_b^{-1} = \mathbf{C}_{-b}$ property made concrete.

6 Software Support: the `convmoment` Package

6.1 Design

We provide an R package, `convmoment`, implementing the rule in R (R Core Team 2024); the package was developed following standard practice for R package structure and documentation (Wickham 2015). The design goal is pedagogical rather than computational: the code is written to look like the rule as taught, so that a student reading the source sees the same three ingredients—binomial coefficients, moments, powers of b—that they used by hand.

```
conv_moment <- function(x, a, k, r) {
  b <- a - k
  mu_from <- c(1, x[1:r]) # mu_0 = 1, followed by moments 1..r
  coeff <- choose(r, 0:r)
  power <- r:0
```

```
  sum(mu_from * coeff * (b^power))
}


conv_moment_all <- function(x, a, k) {
  K <- length(x)
  sapply(1:K, function(r) conv_moment(x, a, k, r))
}


raw2central <- function(raw_moments, origin = 0) {
  mean_val <- raw_moments[1] + origin
  conv_moment_all(raw_moments, a = origin, k = mean_val)
}


central2raw <- function(central_moments, mean_val) {
  conv_moment_all(central_moments, a = mean_val, k = 0)
}
```

The arguments follow the notation of Section 3: `x` is the vector of moments of orders $1, \ldots, K$ about the initial origin `a`, and `k` is the target origin. Note that `central2raw()` expects its input to begin with $\mu_1 = 0$, so that the vector indexing matches the orders.

6.2 Use in Assignments

The package changes what can be asked of students. Because the rule extends to any order at no extra cost, an assignment can require a derivation at an order that appears in no textbook table—order 6, 8, or 10—and students can check their own work with a single call rather than waiting for marked scripts. This closes the feedback loop within the session, and it removes the incentive to pattern-match against a printed answer.

6.3 Verification

We verified the implementation on the sample $X = (20, 25, 29, 32, 40)$ by computing moments of orders 1–5 directly, converting raw to central and central back to raw, and comparing.

The central-to-raw direction reproduces the directly computed raw moments **exactly** in floating point, to the last bit. The raw-to-central direction agrees to a relative error of order 10^{-16}—at the level of machine epsilon—though the *absolute* differences reach roughly 3×10^{-9} at order 5, simply because $\mu'_5 \approx 3.4 \times 10^7$ for these data.

That asymmetry is not a defect, and it is worth showing to students rather than hiding: the central-to-raw direction has $b = \bar{x} > 0$ and sums terms of like sign, while the raw-to-central direction has $b = -\bar{x}$ and therefore alternates, so it loses significant digits to cancellation. The same rule, run in two directions, produces a concrete illustration of catastrophic cancellation. See Section 7.

7 Discussion

7.1 What the Approach Achieves

The gain is not that a difficult result becomes available—the result was never difficult—but that a scattered topic becomes a single one. A student who has learned the origin-shift rule has learned something with the right shape: one operation, one parameter, three familiar special cases, extensible to any order with a tool (Pascal's triangle) they already possess. A student who has learned the textbook formula list has learned five things that will decay independently.

There is a secondary benefit worth naming. Because the rule is derived in three lines from the binomial theorem, it models the habit we want students to have—that a forgotten formula should be re-derived rather than looked up. The formula-list presentation actively teaches the opposite.

7.2 Limitations

The approach does not remove the need for raw moments: something must still supply $\mu'_j(a)$, whether from data, from a generating function, or from a table of known distributional moments. For a distribution whose raw moments are themselves awkward, the origin shift is the easy half of the problem.

Nor does the mnemonic help with conversions that are not origin shifts. Moments to cumulants, in particular, is a different operation with a different combinatorial structure, and students who over-generalize the binomial device to that case will get it wrong. We flag this explicitly in class.

Finally, the evidence and the claim are not at the same level, and we would rather say so than blur it. The argument for level-independence in Section 1.3 is structural: the rule needs only the binomial theorem and linearity, both of which are in place from secondary school onward. The classroom evidence in Section 4 comes from one instructor teaching higher-secondary students, across successive cohorts over six years. Repetition across cohorts is worth something—the observation is not a single good session—but it is repetition by one instructor who designed the approach, which is a specific and well-known source of bias. And it remains the demanding end of the level range rather than a sample of it: it is not a substitute for evidence at undergraduate or graduate level, where students have competing methods available and the comparison would be against generating-function derivations rather than against a formula list.

A proper evaluation would need a comparison group taught conventionally, a delayed retention measure—the claim is about durability, not immediate performance—and an assessment item at an order not covered in instruction. That last point deserves emphasis, since it is the one the present observations most clearly do not address: students here applied the rule at the orders taught (Section 4), which establishes that the rule is learnable in a single session but not that it generalizes in the students' hands to orders they have not seen. Since generativity is precisely what the mnemonic claims over a formula list, that is the first thing a follow-up study should measure. None of it has been done here.

7.3 A Numerical Caution Worth Teaching

The cancellation noted in Section 6 becomes material at high order. Converting uncentered raw moments of large order for data with a large location offset is susceptible to catastrophic cancellation, since the alternating terms are large and nearly equal (Higham 2002); the same concern motivates the specialized updating formulas developed for streaming and parallel computation (Chan et al. 1983; Pébay et al. 2016). For serious high-order work on empirical data, the practical remedies are to center the data first, to use extended-precision arithmetic, or to use a numerically stable one-pass algorithm.

For teaching purposes the point is smaller and more useful: exact algebra and exact floating-point arithmetic are different things, and the origin-shift rule gives a two-line demonstration of the gap. We have found this a more convincing introduction to numerical error than the usual contrived examples, because the students have just derived the formula themselves.

8 Conclusion

We have described a way of teaching moment transformation in which a single origin-shift rule, presented as the symbolic expansion $(a+b)^r$ with $a^j \mapsto \mu'_j(a)$, replaces the list of order-specific formulas found in standard texts. The underlying identity is classical; the contribution is the framing, the reduction of three textbook topics to one parameter $b = a - k$, and an R package that lets students verify derivations at orders no table covers.

In classroom use, students were able to apply the rule at orders beyond those presented in lecture. We report that as informal experience rather than as a measured effect, and a controlled study with a retention component remains to be done.

For instructors wanting to adopt the approach, the minimum viable version is short: derive the rule from the binomial theorem, present Table 1, and give the three cases of b on one slide. That is a single lecture segment, and it replaces material that is otherwise spread across several chapters.

9 Disclosure Statement

The author has no conflicts of interest to declare.

10 Data Availability Statement

The R package `convmoment`, its vignette, and all replication code are publicly available in an open repository. The URLs are withheld here for blind review and will be supplied on acceptance.

11 Supplementary Material

1. **R package `convmoment`:** source code, documentation, and vignette. Available at [URL withheld for blind review].

2. **Replication code:** R script reproducing the numerical examples, the distributional derivations, and the verification results of Section 6.

3. **Teaching slides:** The reveal.js slide deck used to teach this material, covering moments, skewness, and kurtosis, with the origin-shift rule developed through Pascal's triangle; the relevant section is *Changing origin of moments*. The deck is available from the author on request and will be placed in a permanent public archive on acceptance.

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